V4 power and wiring: battery, regulator and schematic

The Mercury V1 sends the servo position signal; a small 2S battery and a 5 V regulator power the SG90. Part sizes below come from the sellers' listings I could reach; I could **not** read the Mercury manual (the Altimeter Cloud site is not reachable from this session), so its servo-header pin order and signal voltage are the items to confirm.

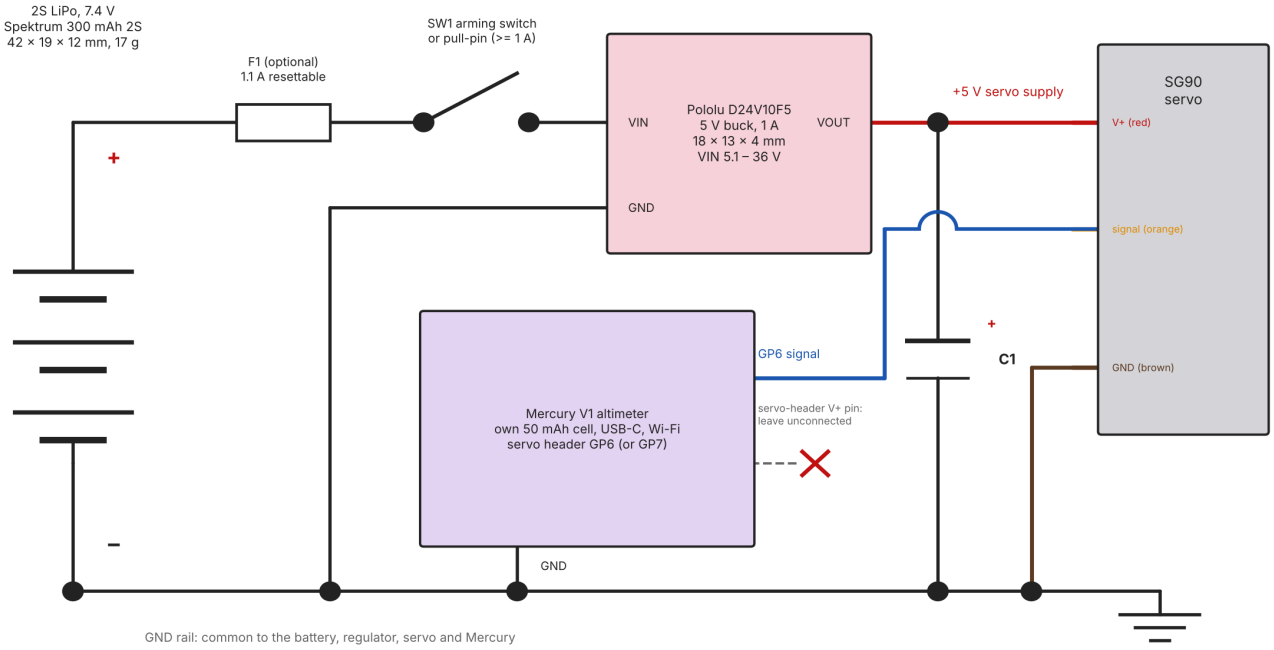
1 Recommended parts

Part	Recommendation	Size / weight (from listings)	Why
Battery	**Spektrum 300 mAh 2S, 30C** (7.4 V nominal, 8.4 V full, PH plug)	42 × 19 × 12 mm, about 17 g (one retailer's listing; measure yours)	Small 2S pack that fits the V4 pocket; 6.0–8.4 V in, which the regulator below accepts; 300 mAh is far more than a flight needs (about 7–12 flights per charge on my estimate)
	Alternatives: Tattu 300 mAh 2S HV; EMAX Tinyhawk S 2S	47 × 17 × 13 mm; 42 × 16.2 × 12.7 mm (listings)	The Tattu is 4.6 mm too long for the new pocket; the EMAX would fit the length and width but is 0.3 mm too tall
Regulator	**Pololu D24V10F5**: 5 V, 1 A step-down	0.5 × 0.7 × 0.14 in (about 13 × 18 × 4 mm), 1 g	Input 5.1–36 V; the SG90 takes 4.8–6 V and draws roughly 0.3–0.55 A at stall (sources vary), so 1 A is about 2× headroom
	Option for more torque: D24V10F6 (6 V, 1 A)	same footprint	SG90 torque rises with voltage (one source: 1.2 → 1.6 kg·cm from 4.8 → 6 V; another lists 1.8 kg·cm at 4.8 V), but this board needs at least 6.1 V in: keep the pack above about 6.5 V
Capacitor C1	470–1000 µF, 10 V, low-ESR electrolytic	about 6.3 mm diameter × 11 mm	Absorbs the servo's current spikes so the regulator and the Mercury do not brown out
Arming	SPST switch ≥ 1 A, or a JST pull-pin / jumper	small slide switch	Keeps the servo dead while you handle it; reach it through the transition ring
Fuse (optional)	1.1 A resettable (polyfuse)	—	Limits a short; the LiPo can deliver far more than the wires should carry
Wire and plugs	24 AWG silicone for power, 28 AWG for the signal; JST-PH 2-pin on the battery (match the plug on the pack you buy)	—	Flexible, heat-tolerant, small

Estimated load: the servo idles near 10 mA and draws about 100–250 mA while moving; one deployment is under a second. A 300 mAh pack therefore lasts many hours of standby (estimate: the regulator's own draw I could not confirm).

2 Wiring schematic

V4 airbrake: servo power and signal (wiring schematic)



C1 = 470 - 1000 uF, 10 V, low-ESR, wired right at the servo pins. The regulator powers the servo; the Mercury only sends the position signal.

Ground is shared. Never join the servo red wire to a Mercury pin; confirm the Mercury servo-header pin order in its manual before wiring.

Servo power and signal. The Mercury has its own 50 mAh battery and does not power the servo.

How it is wired

- **Power path:** battery + → fuse → arming switch → regulator VIN; regulator VOUT (+5 V) → servo red wire and C1; regulator GND, C1 and the servo brown wire all return to the battery –.
- **Signal path:** one Mercury servo header (GP6 or GP7, set in the Mercury configuration as the air-brake servo channel) → servo orange wire. The Mercury GND goes to the same ground rail.
- **The one rule:** the servo's red wire takes power from the regulator only. If the Mercury servo header has a power pin, leave it unconnected (pull the red wire out of the Mercury-side connector shell, or use a servo lead with the red wire cut on that side). Two supplies tied together can damage the Mercury or the regulator.
- **Order of assembly:** bench-test with the servo and a 5 V supply first, then the regulator on the battery, with the switch off until the last step.

3 What I could not verify, and what it means for the CAD

The V4 battery pocket is now sized for the Spektrum 300 mAh 2S. Pocket 42.8 × 19.8 × 12.4 mm (0.4 mm clear per side, 0.2 mm above and below the 42 × 19 × 12 mm pack); the sled's lower block grew from 44.8 × 22.8 to 46.8 × 23.8 mm with its corners trimmed to R25.5 so they clear the opened base ring. The housing stays 94.3 mm: the extra height came out of the 3.3 mm floor between the two layers, which is now 1.8 mm thick. **Battery strap.** The pocket is open at the bottom, so a belly band holds the pack: a nylon tie or narrow hook-and-loop strap (up to 4 mm wide, 1.5 mm thick, about 90 mm long) runs under the pack, up the outside of the lower sled block and across a 1 mm groove in its top. Two channels are modelled, at x = ±17.0…21.4 mm (between the upper block and the end wall); use one or both. The small ribs are notched there, and the transition ring has two notches (6.3 mm × 29 mm, 1.8 mm deep) so the band has room under the sled. I checked a 3.4 mm band in the model: it does not touch the sled, the transition ring or the housing. Still to do on your side: **measure the real pack** (the 42 × 19 × 12 mm figure is one retailer's listing) and allow for the plug and the 3-pin balance lead (fold them beside or on the pack).

- **Mercury servo header:** the Altimeter Cloud flight page I found says the two onboard servo headers are GP6 and GP7, but I could not read the pin order or the signal voltage. Check its manual (the "Mercury V1 schematic" page). The SG90 usually accepts a 3.3 V signal when powered at 5 V, but test it on the bench before flying.
- **SG90 figures vary by source:** stall current 0.3 to 0.55 A, torque 1.2 to 1.8 kg·cm at 4.8 V depending on the version. The V3/V4 torque check used 0.176 N·m (1.8 kg·cm) at 5 V; measure your own servo.
- **Regulator input:** Pololu warns of voltage spikes above its 36 V maximum with long leads at high input voltages; with a 2S pack (8.4 V full) this is not a concern. The D24V10F5 needs at least 5.1 V in, so stop flying a 2S pack that has sagged near 6 V.
- **LiPo safety:** the Spektrum is a standard LiPo (charge to 4.2 V per cell; an HV pack such as the Tattu needs a 4.35 V setting); do not discharge below about 3.3 V per cell; keep the pack away from the ejection-charge heat and from crushing; use a fire-safe bag for charging.
- **Altimeter pressure:** the Mercury reads static pressure, so the bay needs vent holes to the outside (not modelled in the CAD).

Sources

- Altimeter Cloud flight page (GP6/GP7 servo headers, air-brake channel) and the Mercury V1 schematic page (not readable from this session).
- Pololu D24V10F5 5 V, 1 A step-down regulator and D24V10F6 6 V version.
- Spektrum 300 mAh 2S listings: Readymade RC. Other packs considered: Tattu 300 mAh 2S listings: GetFPV, Aloft Hobbies, Gens Ace; Turnigy nano-tech 2S 300 mAh: Botland.
- SG90 figures: SG90 datasheet copy and ProtoSupplies.